



Medical Imaging Evaluation Handbook

(For X-ray Fracture Detection & Beyond)



0. First Principle (Don't Skip This)

Every metric answers **one of three questions**:

- 1. **Discrimination** → Can the model separate classes?
- 2. **Calibration** → Are predicted probabilities trustworthy?
- 3. **Localization** → Is the model looking at the correct region?

If your evaluation doesn't cover all three → incomplete.



1. Problem Types (Metrics depend on this)

Task	Example	Metrics
Classification	Fracture vs Normal	Accuracy, Recall, AUC
Detection	Find fracture location (box)	mAP, IoU
Segmentation	Pixel-wise fracture region	Dice, IoU
Multi-label	Multiple conditions	Macro/micro metrics



2. Confusion Matrix (Core of Everything)

	Predicted Positive	Predicted Negative
Actual Positive	TP	FN
Actual Negative	FP	TN



Interpretation

- TP → Correct detection
- FN → Missed disease (**critical**)
- FP → False alarm
- TN → Correct rejection

👉 In medicine: **FN is usually the worst**



3. Classification Metrics (Complete Set)



Accuracy

$$\frac{TP + TN}{Total}$$

- Misleading in imbalanced datasets

Precision (Positive Predictive Value)

$$\frac{TP}{TP + FP}$$

👉 “When model says fracture, how often correct?”

Recall (Sensitivity)

$$\frac{TP}{TP + FN}$$

👉 “How many real fractures did we detect?”

🔥 Most critical in diagnosis

Specificity

$$\frac{TN}{TN + FP}$$

👉 “How well we avoid false alarms”

F1 Score

$$\frac{2PR}{P + R}$$

👉 Balance of precision & recall

Balanced Accuracy

$$\frac{Sensitivity + Specificity}{2}$$

👉 Useful for imbalanced data



MCC (Matthews Correlation Coefficient)

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

👉 Best single metric for imbalance

- +1 = perfect
- 0 = random



Cohen's Kappa

Measures agreement beyond chance

👉 Useful when comparing with radiologists



4. Threshold-Based vs Threshold-Free Metrics

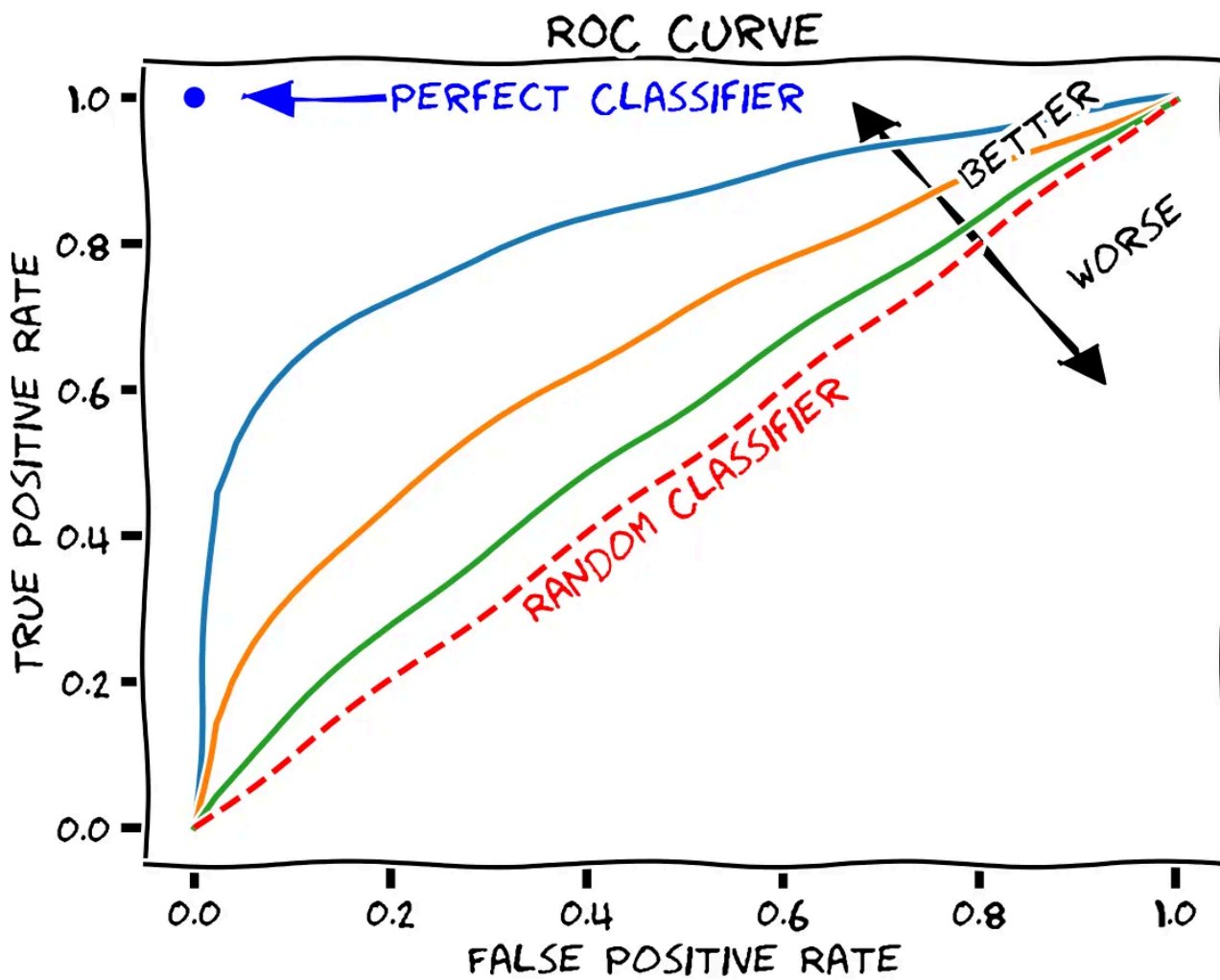


ROC Curve & AUC

- Measures **ranking ability**

Interpretation:

- AUC = probability model ranks positive higher than negative



error code: 1020



Precision-Recall Curve

Better when:

- Dataset is imbalanced (which yours is)

Key Difference

Metric	Use Case
ROC-AUC	Balanced datasets
PR-AUC	Imbalanced datasets

5. Calibration Metrics (Almost Everyone Ignores This)

This is where most “high accuracy models” fail clinically.

What is Calibration?

If model says:

- “80% fracture”

 Is it actually correct 80% of the time?

Metrics

◆ Brier Score

$$\text{Mean}(\text{predicted} - \text{actual})^2$$

- Lower is better

◆ Expected Calibration Error (ECE)

Measures mismatch between:

- predicted probability vs actual outcome

◆ Reliability Diagram

 Image

 Image

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Interpretation

Pattern	Meaning
Below diagonal	Overconfident
Above diagonal	Underconfident

6. Detection Metrics (Bounding Boxes)

Used if you localize fractures.

IoU (Intersection over Union)

$$\frac{Overlap}{Union}$$

mAP (Mean Average Precision)

- Measures detection performance across thresholds

Interpretation

- mAP@0.5 → loose localization
- mAP@0.75 → strict

7. Segmentation Metrics (Pixel-Level)

Dice Coefficient

$$\frac{2TP}{2TP + FP + FN}$$

👉 Most used in medical imaging

IoU (Jaccard Index)

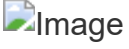
$$\frac{TP}{TP + FP + FN}$$

Difference

- Dice is more forgiving
- IoU is stricter



Segmentation Example



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8. Localization Metrics (Weak Supervision / Grad-CAM)



Pointing Game

- Check if peak activation lies inside fracture region



Localization Accuracy

- % of heatmaps overlapping annotated region



Energy-Based Metrics

- How much heatmap mass lies inside ground truth

9. Explainability Metrics (Grad-CAM & Beyond)

What to Evaluate

1. Faithfulness

- Does heatmap reflect model decision?

2. Localization

- Does it align with pathology?

3. Robustness

- Does explanation change drastically with small noise?

Reality

- Grad-CAM is **not reliable alone**
- Compare with:
 - Grad-CAM++
 - Score-CAM
 - Eigen-CAM

10. Statistical Evaluation (You NEED this for PhD level)



Confidence Intervals

- Report uncertainty in metrics



Bootstrapping

- Resample dataset → compute metric distribution



Hypothesis Testing

- Compare models (e.g., AUC difference)



11. Error Analysis Metrics



Per-Class Metrics

- Fracture types (wrist, hip, etc.)



Hard Case Analysis

- Small fractures
- Low contrast
- Occlusions

Subgroup Analysis

- Age groups
- Imaging views

12. Training Metrics (Model Behavior)

Loss Curve

- Optimization progress

Accuracy Curve

- Learning behavior

Red Flags

Pattern	Problem
Train $\uparrow$ Val $\downarrow$	Overfitting
Both low	Underfitting

13. Clinical Metrics (Rarely Used but Powerful)

NPV (Negative Predictive Value)

$$\frac{TN}{TN + FN}$$

👉 “If model says normal, can we trust it?”

PPV (Same as Precision)

Decision Curve Analysis

- Measures clinical usefulness

14. Generalization Metrics

Cross-Dataset Performance

Train → MURA

Test → FracAtlas

👉 Measures robustness

Domain Shift Sensitivity

- Performance drop across datasets

15. What a Complete Evaluation Looks Like

✓ MUST HAVE

- Confusion matrix metrics
- ROC + PR curves
- Calibration analysis
- Error breakdown

✓ STRONG PAPER

- Localization metrics
- Explainability validation
- Cross-dataset testing
- Statistical significance

✓ TOP-TIER RESEARCH

- Clinical validation
- Human vs AI comparison
- Failure mode taxonomy



Final Reality Check

If your evaluation:

- Ignores calibration → not clinically usable
- Ignores FN → dangerous
- Ignores localization → not trustworthy
- Ignores generalization → not publishable



Bottom Line

- **Metrics ≠ understanding**
- Each metric answers a different failure mode
- Real research = connecting them into a **coherent story**